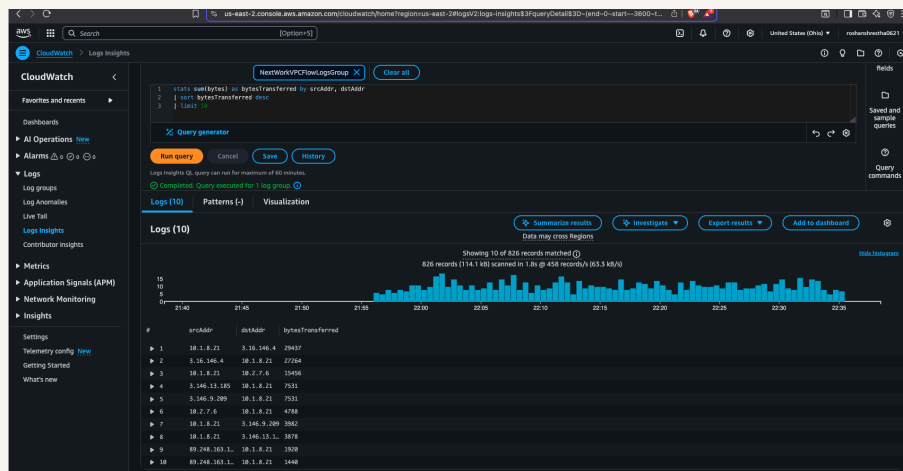




VPC Monitoring with Flow Logs



Roshan Shrestha





Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated network in the amazon cloudspace. It is useful because it resembles a traditional network that we would operate in our own data center.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create two EC2 instances, one in each VPC. Open a peering connection between the two VPCs. Set up Flow logs using CloudWatch. Test ping messages between the two instances and analyze the traffic using flow log

One thing I didn't expect in this project was...

One thing I didn't expect in this project was being able to query into the flow logs.

This project took me...

This project took me about 2 hrs.



In the first part of my project...

Step 1 - Set up VPCs

In this step, we are creating two VPCs from scratch which we will use to setup a peering network connection between them.

Step 2 - Launch EC2 instances

In this step, we are launching an EC2 instance in each VPC's public subnet to test a peering connection.

Step 3 - Set up Logs

In this step, we are setting up a way to track all inbound and outbound network traffic and also setting up a space that stores all of these records.

Step 4 - Set IAM permissions for Logs

In this step, we are going to give VPC Flow Logs the permission to write logs and send them to CloudWatch.



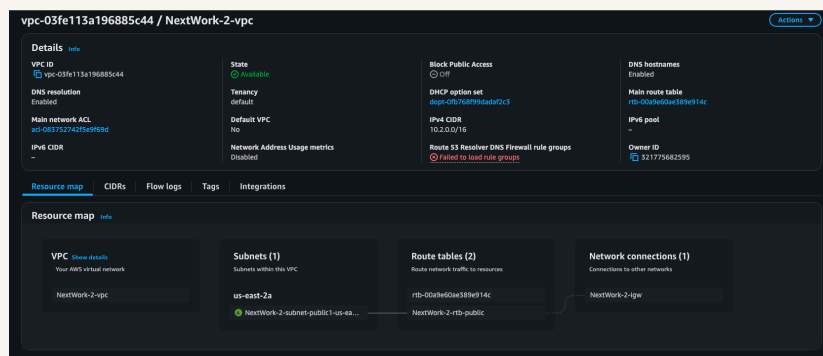
Multi-VPC Architecture

I started my project by launching two VPCs. I created two public subnets - one for each VPC.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because we don't want any overlapping IP addresses which result in routing problems and connectivity issues.

I also launched EC2 instances in each subnet

My EC2 instances security groups allow all inbound SSH and ICMP - IPv4 traffic from all sources and allow all outbound traffic. This is because we need the ICMP - IPv4 to be able to ping from one instance to another and to the public internet.

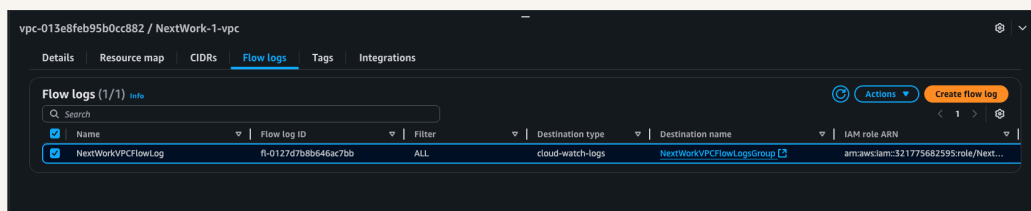




Logs

Logs are records of everything happening with our VPCs.

Log groups are a big folder in AWS where you keep related logs together. They are region specific.





IAM Policy and Roles

I created an IAM policy because VPC Flow Logs by default don't have the permission to record logs and store them in your CloudWatch log group. This policy makes sure that your VPC can now send log data to your log group.

I also created an IAM role because we cannot assign IAM policies directly to resources. Policies are combined under a role and that role is assigned to a resource.

A custom trust policy is a specific type of policy, different from IAM policies. While IAM policies help you define the actions a user/service can or cannot do, custom trust policies are used to very narrowly define who can use a role.



Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "Statement1",  
6       "Effect": "Allow",  
7       "Principal": {  
8         "Service": "vpc-flow-logs.amazonaws.com"  
9       },  
10      "Action": "sts:AssumeRole"  
11    }  
12  ]  
13 }
```



In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step we are going to setup VPC peering and send test messages from EC2 instance 1 to 2 because we want to generate some network traffic and see whether our flow logs can pick up on them.

Step 6 - Set up a peering connection

In this step we are creating a peering connection to create a direct route between VPCs 1 and 2.

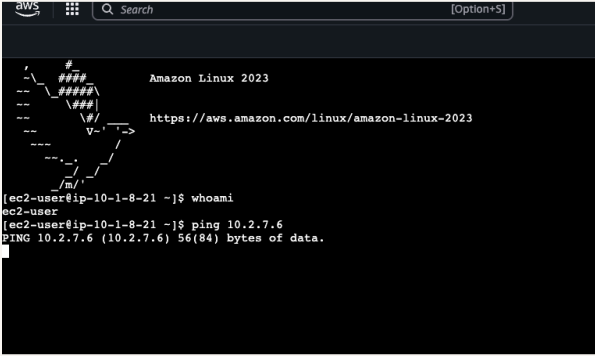
Step 7 - Analyze flow logs

In this step we are reviewing the flow logs recorded about VPC 1's public subnet and analyzing the logs.



Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means the two instances are not able to communicate with each other. The request from instance 1 is not reaching instance 2.



```
aws
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-1-8-21 ~]$ whoami
ec2-user
[ec2-user@ip-10-1-8-21 ~]$ ping 10.2.7.6
PING 10.2.7.6 (10.2.7.6) 56(84) bytes of data.
```

I could receive ping replies if I ran the ping test using the other instance's public IP address, which means the two instances are able to communicate using the public IP which uses the public internet.



Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because we don't have a direct route to VPC2 from VPC 1.

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that both VPCs can directly communicate to each other using their private addresses utilizing the peering connection VPC 1 <> VPC 2.

rtb-04617046be888a1e / NextWork-1-rtb-public			
Details	Routes	Subnet associations	Edge associations
Route propagation			
Tags			
Routes (3)			
Filter routes			
Destination	Target	Status	Propagated
0.0.0.0/0	igw-05d96e9670f9c7e6d	Active	No
10.1.0.0/16	local	Active	No
10.2.0.0/16	pcx-04c72a14ab8c4da9c	Active	No



Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means that the two instances are successfully using the peering connection we setup VPC 1 <> VPC 2.

```
aws
[Option+S]
C
--- 10.2.7.6 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4184ms

[ec2-user@ip-10-1-8-21 ~]$ ping 10.2.7.6
PING 10.2.7.6 (10.2.7.6) 56(84) bytes of data:
64 bytes from 10.2.7.6: icmp_seq=1 ttl=127 time=0.621 ms
64 bytes from 10.2.7.6: icmp_seq=2 ttl=127 time=0.198 ms
64 bytes from 10.2.7.6: icmp_seq=3 ttl=127 time=0.249 ms
64 bytes from 10.2.7.6: icmp_seq=4 ttl=127 time=0.209 ms
64 bytes from 10.2.7.6: icmp_seq=5 ttl=127 time=0.218 ms
64 bytes from 10.2.7.6: icmp_seq=6 ttl=127 time=0.208 ms
64 bytes from 10.2.7.6: icmp_seq=7 ttl=127 time=0.229 ms
64 bytes from 10.2.7.6: icmp_seq=8 ttl=127 time=0.202 ms
64 bytes from 10.2.7.6: icmp_seq=9 ttl=127 time=0.204 ms
64 bytes from 10.2.7.6: icmp_seq=10 ttl=127 time=0.194 ms
64 bytes from 10.2.7.6: icmp_seq=11 ttl=127 time=0.202 ms
64 bytes from 10.2.7.6: icmp_seq=12 ttl=127 time=0.214 ms
64 bytes from 10.2.7.6: icmp_seq=13 ttl=127 time=0.218 ms
64 bytes from 10.2.7.6: icmp_seq=14 ttl=127 time=0.212 ms
64 bytes from 10.2.7.6: icmp_seq=15 ttl=127 time=0.193 ms
64 bytes from 10.2.7.6: icmp_seq=16 ttl=127 time=0.200 ms
64 bytes from 10.2.7.6: icmp_seq=17 ttl=127 time=0.202 ms
64 bytes from 10.2.7.6: icmp_seq=18 ttl=127 time=0.206 ms
64 bytes from 10.2.7.6: icmp_seq=19 ttl=127 time=0.210 ms
64 bytes from 10.2.7.6: icmp_seq=20 ttl=127 time=0.189 ms
64 bytes from 10.2.7.6: icmp_seq=21 ttl=127 time=0.234 ms
64 bytes from 10.2.7.6: icmp_seq=22 ttl=127 time=0.201 ms
64 bytes from 10.2.7.6: icmp_seq=23 ttl=127 time=0.184 ms
64 bytes from 10.2.7.6: icmp_seq=24 ttl=127 time=0.214 ms
64 bytes from 10.2.7.6: icmp_seq=25 ttl=127 time=0.204 ms
64 bytes from 10.2.7.6: icmp_seq=26 ttl=127 time=0.188 ms
64 bytes from 10.2.7.6: icmp_seq=27 ttl=127 time=0.192 ms
64 bytes from 10.2.7.6: icmp_seq=28 ttl=127 time=0.192 ms
64 bytes from 10.2.7.6: icmp_seq=29 ttl=127 time=0.226 ms
64 bytes from 10.2.7.6: icmp_seq=30 ttl=127 time=0.208 ms
64 bytes from 10.2.7.6: icmp_seq=31 ttl=127 time=0.196 ms
64 bytes from 10.2.7.6: icmp_seq=32 ttl=127 time=0.205 ms
64 bytes from 10.2.7.6: icmp_seq=33 ttl=127 time=0.204 ms
64 bytes from 10.2.7.6: icmp_seq=34 ttl=127 time=0.186 ms
64 bytes from 10.2.7.6: icmp_seq=35 ttl=127 time=0.197 ms
64 bytes from 10.2.7.6: icmp_seq=36 ttl=127 time=0.194 ms
64 bytes from 10.2.7.6: icmp_seq=37 ttl=127 time=0.186 ms
64 bytes from 10.2.7.6: icmp_seq=38 ttl=127 time=0.214 ms

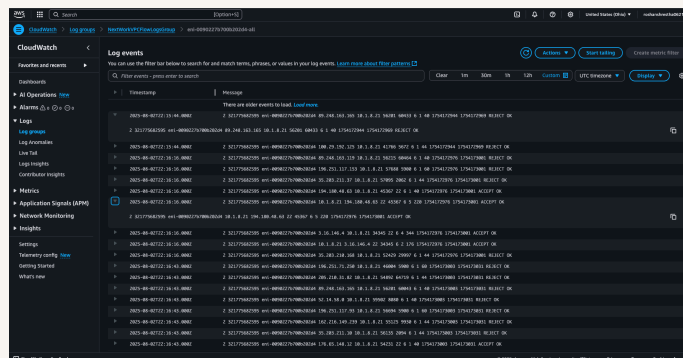
i-02986a9c585a5153b (Instance - NextWork VPC 1)
PublicIPs: 5.128.205.18 - PrivateIPs: 10.1.8.21
```



Analyzing flow logs

Flow logs tell us about the version of the flow log, the AWS account that created it, network interface, source and destination IPs and ports, no. of data packets and bytes, start/end time, if the traffic was allowed in and if logging was successful

The flow log I've captured tells us about the different sources that pinged the EC2 instance 2 and if the ping was successful.

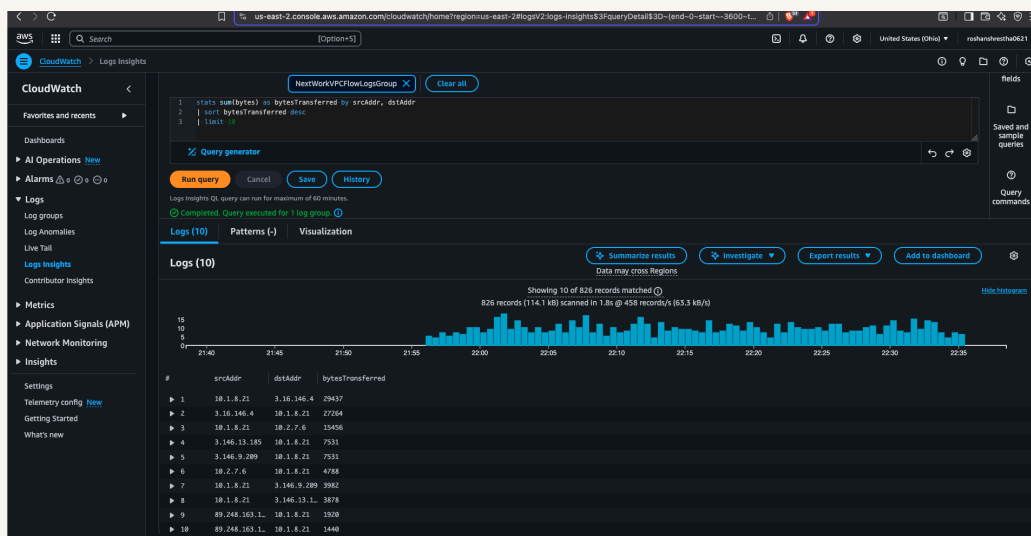




Logs Insights

Logs Insights is a CloudWatch feature that analyzes your logs. In Log Insights, you use queries to filter, process and combine data to help you troubleshoot problems or better understand your network traffic.

I ran the query "Top 10 byte transfers by source and destination IP addresses". This query analyzes the most data transferred between source and destination IP addresses.





nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects

